

AUTOMATED AUDIO CAPTIONING WITH ENHANCED ENCODER ARCHITECTURES AND LLM-BASED POST-PROCESSING

Technical Report

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ABSTRACT

Automated Audio Captioning (AAC) involves generating natural-language descriptions of audio clips, a task requiring systems to capture present sound events, their relationships, and the broader acoustic scene. In this technical report, we present our implementation for Task 6 of the DCASE 2024 Challenge. Our work builds upon advancements in the field—specifically, systems leveraging multi-encoder fusion and Large Language Model (LLM) enriched summarization [1]. Because open-source implementations for these complex post-processing and multi-level fusion pipelines were previously unavailable, we first establish a complete, fully configurable implementation of this pipeline from scratch to serve as our baseline.

Starting from this replicated configuration, we expand the architecture in three key directions: 1) we introduce a third audio encoder, the Audio Spectrogram Transformer (AST), alongside the existing BEATs and ConvNeXt branches 2) we propose a learned cross-attention resampling strategy for encoder fusion and 3) we implement a pre-training regimen. Our pipeline refines the fused acoustic representations using a Conformer post-encoder before passing them to a BART language decoder. During inference, we bypass standard beam search in favor of a multi-stage approach: nucleus sampling generates diverse candidate captions, a CLAP model filters them based on audio-text similarity, and an LLM consolidates the top candidates into a single fluent caption.

Index Terms— Automated audio captioning, Clotho, DCASE 2024, FENSE, Fusion, LLMs, Nucleus Sampling, Reranking

1. INTRODUCTION

Automated Audio Captioning (AAC) is the task of generating a natural-language description of the content of an audio clip. Unlike audio tagging or sound event detection, which map an input signal to a fixed set of categorical labels, AAC requires a system to produce a free-form sentence describing the sound events present, their relationships, and the broader acoustic scene implied by the recording. For example, a model should be able to turn a recording of footsteps, traffic, and speech into a caption such as “people are walking and talking near a busy street.” AAC therefore sits at the intersection of machine listening and natural language generation, and is the focus of Task 6 of the DCASE 2024 Challenge [2].

Despite recent progress, AAC remains difficult. Generated captions must balance acoustic accuracy with linguistic fluency: they should correctly identify the sound events while remaining grammatical and natural-sounding. Furthermore, available captioned au-

dio datasets are small compared to those used in image captioning or speech recognition, which limits the capacity of models trained from scratch and makes generalization a persistent challenge. The official DCASE 2024 Task 6 baseline, CoNeTTE [3], addresses this using a frozen ConvNeXt-Tiny encoder and a lightweight Transformer decoder, but leaves significant headroom for improvement.

Top-performing submissions to the DCASE 2024 Task 6 challenge improved upon this baseline through two complementary directions: acquiring richer audio representations via multi-encoder fusion, and integrating Large Language Models (LLMs) at inference time to re-rank and summarize candidate captions. Notably, the winning system by Jung *et al.* [1] demonstrated that combining a BEATs [4] and ConvNeXt [5] encoder with an LLM-enriched summarization pipeline drastically improves caption quality. However, open-source implementations for this complex multi-stage pipeline were previously unavailable, creating a barrier to further research.

To address this gap, our work first establishes a complete, open-source replication of this multi-encoder fusion and LLM post-processing pipeline from scratch, serving as a foundation. Building upon this baseline, we expand the architecture to capture more acoustic representations and potentially better cross-modal alignment. Specifically, we extend the state-of-the-art by: 1) Introducing the Audio Spectrogram Transformer (AST) [6] as a third parallel encoder branch alongside BEATs and ConvNeXt. 2) Proposing a new learned cross-attention resampling strategy to fuse these disparate encoder representations more effectively than standard concatenation. 3) Implementing a pre-training regime using AudioCaps and MixUp augmentation before fine-tuning on Clotho.

We evaluate our systems on Clotho v2 [7] using FENSE [8] as the primary metric. Our code is publicly available at <https://github.com/RobertTylman/Automatic-Audio-Captioning-ML-2026>.

2. RELATED WORK

2.1. The CoNeTTE Baseline

The official DCASE 2024 Task 6 baseline is based on CoNeTTE [3]. CoNeTTE uses a ConvNeXt-Tiny encoder adapted for audio classification and pretrained on AudioSet [5]. During caption training, the encoder is kept frozen and produces frame-level audio embeddings. These embeddings are passed through a lightweight projection module, consisting of dropout, a linear layer, ReLU activation, and dropout, before being given to the caption decoder.

The decoder is a vanilla Transformer [9] trained from scratch

with cross-entropy loss and teacher forcing. At inference time, captions are generated using beam search. CoNeTTE is trained on several audio-captioning datasets, including Clotho, AudioCaps, WavCaps, and MACS, and uses AdamW optimization, label smoothing, and Mixup-style augmentation on the audio embeddings. Because it is compact, well documented, and modular, CoNeTTE provides a strong starting point for studying extensions to the encoder, decoder, and inference pipeline.

2.2. LLM-Based Audio Caption Decoding

Chen *et al.* [10] propose the SJTU-THU system, which replaces the lightweight Transformer decoder with a 7B-parameter Vicuna large language model. Audio is encoded with the Efficient Audio Transformer (EAT), and a two-layer linear projector downsamples the EAT output before injecting it into the LLM as pseudo-text tokens. The system is adapted using LoRA: only the LoRA parameters on the query and value projections of Vicuna are fine-tuned, while the audio encoder and most of the LLM remain frozen.

At inference time, the system generates multiple candidate captions with beam search and then uses a frozen CLAP model [11] to re-rank candidates by audio-text cosine similarity. This work is important because it demonstrates that an LLM can be used directly as an AAC decoder, rather than only as a post-processing module. It also shows that CLAP-based filtering is an effective inference-time strategy for selecting acoustically relevant captions.

2.3. Encoder Fusion, LLM-Enriched Summarization and Ensembling Strategies

Jung *et al.* [1] reported the highest-FENSE submissions to DCASE 2024 Task 6, achieving a peak FENSE score of 0.544 compared to the 0.504 baseline. Their core system utilizes BEATs [4] and ConvNeXt [5] as parallel frozen audio encoders. Hidden layers from these encoders are aggregated via a “concatenate-then-compress” strategy and combined through early or late fusion, before being decoded using a BART model [12].

Crucially, their top performance relied heavily on robust ensembling and a multi-stage inference pipeline. Rather than generating a single caption, their system uses nucleus sampling to generate 64 candidates, filters them using CLAP audio-text similarity, and employs a GPT-4-Turbo prompt to summarize the surviving candidates into a single fluent caption. To maximize performance, they employed massive ensemble strategies across multiple architectures, utilizing k-fold cross-validation to perform both model-level ensembling (averaging weights) and caption-level ensembling (summarizing the sampled captions from multiple distinct models simultaneously).

3. PROPOSED METHODS

We propose a modular multi-encoder fusion architecture for AAC that builds upon the baseline established by winning submissions [1]. Our system extends this baseline by introducing an additional pretrained audio encoder and a new learned cross-attention fusion module. Figure 1 shows an overview of the proposed system.

Each component of the architecture is described below. All architectural choices are controlled through YAML configuration files, allowing encoder combinations and fusion strategies to be compared without modifying code.

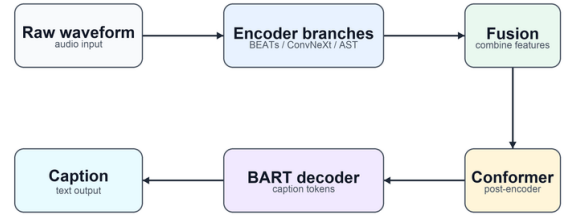


Figure 1: Overview of the proposed automated audio captioning pipeline. The raw waveform is processed by multiple pretrained encoder branches, fused into a shared representation, refined by a Conformer post-encoder, and decoded into a natural-language caption using BART.

3.1. Encoder Branches

Our system supports three pretrained audio encoder branches. **BEATs** [4] is used as the primary anchor encoder and is always active, producing strong frame-level audio embeddings through iteratively self-supervised learning. **ConvNeXt** [5] is included as a secondary convolutional encoder adapted from computer vision and pretrained on AudioSet.

To expand the richness of the acoustic representation, we introduce a third optional branch: the **Audio Spectrogram Transformer (AST)** [6]. AST treats the log-mel spectrogram as a sequence of overlapping patches and processes them entirely with multi-head self-attention, giving it a larger global receptive field than convolutional models. Each encoder internally handles its own resampling and feature preparation.

3.2. Fusion Module

A critical challenge in multi-encoder AAC is combining branches that produce sequences of different temporal lengths. Jung *et al.* [1] found that early fusion in the feature dimension yielded the best performance. We implemented this as our starting point, but subsequently developed two alternative strategies to address its limitations.

1. Feature Fusion (Baseline): This method aligns the time axes of all secondary branches to match the BEATs time resolution using 1D linear interpolation. The aligned feature vectors are then concatenated along the feature dimension and compressed via a multi-layer perceptron (MLP). While effective, this approach assumes that latent acoustic features can be linearly stretched or compressed, a flawed assumption, as complex acoustic events do not interpolate linearly in the latent space.

2. Learned Cross-Attention Resampling (Proposed): To overcome the limitations of linear interpolation, we propose a learned alignment mechanism. In this mode, the BEATs sequence defines the output time axis and acts as the *query* in a Multi-Head Cross-Attention module. The secondary branches retain their native sequence lengths and act as the *keys* and *values*. This allows the model to selectively attend to the most relevant acoustic features from the secondary encoders dynamically, rather than blindly squashing them into a fixed time axis.

3. Sequence Fusion: Instead of attempting to align the time axes, this strategy projects each branch to a shared hidden dimension and concatenates them end-to-end along the temporal axis. This produces a much longer sequence, relying entirely on the downstream Conformer and decoder cross-attention to resolve the temporal relationships.

3.3. Conformer Post-Encoder and Caption Decoder

After fusion, the combined representation is passed through a Conformer post-encoder [13], which interleaves multi-head self-attention with convolution modules to model both global and local acoustic structures. The refined sequence is then passed to a BART sequence-to-sequence decoder [12], which translates the acoustic tokens into natural language.

3.4. Data Augmentation and Pre-Training

To combat the inherent data scarcity of the Clotho dataset, we explored two distinct data expansion strategies:

- **Stable Audio Augmentation:** We utilized a synthetic data pipeline where two synthetic audio clips were generated for every ground-truth Clotho caption using Stable Audio. To ensure quality, we filtered the synthetic audios using a CLAP model, retaining them only if their audio-text similarity score was lower than the original real audio’s score, preventing hyper-exaggerated synthetic alignments from skewing the natural data distribution.
- **Pre-training Regimen:** We implemented a two-stage training process. Models were first pre-trained on AudioCaps [14] and a Clotho-MixUp dataset (generated by blending audio waveforms and merging their captions), before being fine-tuned exclusively on Clotho v2.

3.5. Extended Inference Pipeline

Beyond the core training and validation pipeline, we implement an extended inference stage designed to improve caption quality at test time without retraining the model. This stage is inspired by the inference pipeline of Jung *et al.* [1], which reported the highest FENSE scores of any DCASE 2024 Task 6 submission.

The extended pipeline proceeds in three steps. First, nucleus (top- p) sampling [15] replaces greedy decoding to generate a diverse set of candidate captions for each audio clip. Sampling from the nucleus of the probability distribution encourages the decoder to explore multiple plausible descriptions instead of collapsing to the single most probable token at each step.

Second, a frozen CLAP model [11] scores each candidate caption by computing the cosine similarity between the audio embedding and the caption text embedding in CLAP’s joint audio-language space. Low-scoring candidates are discarded.

Third, an LLM consolidates the surviving candidates into a single fluent caption. The goal is to merge complementary content from the strongest candidates while suppressing hallucinated or inconsistent details. This inference extension is modular and does not require retraining any component of the captioning model.

3.6. Model Configurations

All architectural choices are specified through YAML configuration files. Across all runs, the BART decoder uses the

facebook/bart-base pretrained weights with model dimension $d_{\text{model}} = 768$, 6 decoder layers, 12 attention heads, feedforward dimension 3,072, dropout 0.1, and label smoothing 0.1. The audio projection between the fusion module and the decoder is a learned linear layer with layer normalization, no bias, and weights initialized from $\mathcal{N}(0, 10^{-3})$. The Conformer post-encoder has input dimension 768, 8 attention heads, feedforward dimension 1,536, 2 layers, depthwise convolution kernel size 31, and dropout 0.1. The fusion output dimension is 768, with dropout 0.1 and 8 resampling heads for learned cross-attention fusion. The specifications of each different configuration is documented in Table 1.

3.7. Optimizer and Learning Rate Schedule

All models are trained with AdamW [16] with weight decay 10^{-2} , applied only to trainable parameters. Runs without a scheduler which are the pretraining, use a fixed learning rate of 2×10^{-4} . Runs with a scheduler which are finetuning or only training on clotho-v2 use an initial learning rate of 1×10^{-4} with a custom *InverseLR-HalfMix* schedule:

$$\eta_t = \eta_0 \cdot (1 - \rho^{t+1}) \cdot \left(\frac{1}{2} \left(1 + \frac{t}{\gamma} \right)^{-p} + \frac{1}{2} \right) \quad (1)$$

where $\rho = 0.99$ controls exponential warmup, $\gamma = 2000$ is the inverse decay timescale, and $p = 10$ is the decay power. The first factor provides a smooth warmup over the early steps; the second factor decays like an inverse power law but asymptotes to $0.5 \eta_0$ rather than zero, preventing the learning rate from collapsing over long training runs. The scheduler is applied per step.

3.8. Inference Pipeline

At evaluation time two generation modes are supported. In **greedy** mode, the decoder generates a single caption per audio clip using greedy decoding with a 3-gram no-repeat constraint and a maximum length of 32 tokens. In **rerank** mode, nucleus sampling with temperature 0.5 and $p = 0.95$ generates 64 candidate captions per clip with the same no-repeat constraint. Candidates are then scored by a hybrid objective combining CLAP [11] audio-text cosine similarity (weight 0.7) and normalized decoder negative log-likelihood (weight 0.3), and deduplicated. The top 32 candidates by hybrid score are retained, then further reduced to 12 by pair-wise centrality selection in CLAP text-embedding space. Finally, the 12 selected captions are passed to GPT-4.1-mini with a Caption Summarization-Activating Prompt (SAP), which instructs the model to consolidate the candidates into a single fluent sentence of approximately ten words in present-continuous tense. If the LLM call fails, the top-ranked candidate is returned as a fallback.

3.9. Metrics

The primary evaluation metric is FENSE [8], which combines a Sentence-BERT cosine similarity score with a fluency penalty that down-weights grammatically disfluent captions. FENSE is the ranking metric for the DCASE 2024 Task 6 leaderboard and is robust to paraphrasing.

3.10. Pre-Training

As an additional configuration, we experimented with pre-training the model on AudioCaps and MixUp of Clotho, reaching a good

Data Expansions			Encoders			Fusion Mode	FENSE	
StableAudio	MixUp	AudioCaps	BEATs	ConvNext	AST		Greedy	Rerank
✗	✗	✗	✓	✓	✗	Feature	0.430	0.515
✗	✗	✗	✓	✓	✗	Learned Res.	0.464	0.512
✗	✗	✗	✓	✓	✗	Sequence	0.466	0.518
✗	✗	✗	✓	✓	✓	Feature	0.449	0.517
✗	✗	✗	✓	✓	✓	Learned Res.	0.445	0.504
✗	✗	✗	✓	✓	✓	Sequence	0.457	0.506
✓	✗	✗	✓	✓	✓	Feature	0.449	0.508
✓	✗	✗	✓	✓	✓	Learned Res.	0.444	0.502
✓	✗	✗	✓	✓	✓	Sequence	0.458	0.514
✗	Pretrain	Pretrain	✓	✓	✗	Learned Res.	0.471	0.525
✗	Pretrain	Pretrain	✓	✓	✓	Learned Res.	0.477	0.523
✗	Pretrain	Pretrain	✓	✓	✓	Sequence	0.486	0.531

Table 1: Summary of evaluated AAC configurations with expanded data sources.

validation loss on that, and use the checkpoint for fine-tuning the model using the Clotho dataset. All pretraining schemes were performed with a larger learning rate of 2×10^{-4} .

4. RESULTS

Table 1 summarizes the evaluated model configurations, including the pre-training datasets, encoder configurations, fusion mode, and FENSE scores.

The result’s table is divided to three sections:

- Winning paper re-implementation using the different fusion modes.
- Adding a third encoder, scoring each greedy and reranked LLM results.
- Pretrained model using MixUp and AudioCaps as a pretrained checkpoint, and then fine-tuned on the Clotho dataset.

4.1. Overfitting and the Nature of LLM Reranking

When interpreting these results, it is important to understand the constraints of the data. Clotho v2 is a relatively small dataset, throughout our un-pretrained experiments, models overfit rapidly, resulting in climbing validation losses. Consequently, the performance ceiling was often dictated by data scarcity rather than architectural limits.

Furthermore, the post-processing pipeline introduces noise. Because GPT-based summarization is non-deterministic, reranked FENSE scores naturally fluctuate. For instance, a greedy score jump from 0.430 to 0.449 is a mathematically significant improvement in raw acoustic translation, but after LLM summarization, their reranked scores (0.515 and 0.517) appear nearly identical. Therefore, we primarily rely on *greedy FENSE* to assess the true alignment quality of the underlying acoustic models, while the reranked scores demonstrate the absolute upper bound achievable through inference-time generation.

4.2. Evaluating Encoder Richness and Fusion

Our baseline implementation of 2-encoder Feature Fusion (1D linear interpolation) achieved a greedy score of 0.430. Upgrading this to our proposed **Learned Resampling** yielded a jump to 0.464, confirming that cross-attention successfully aligns complex acoustic spaces where simple linear interpolation fails.

Similarly, expanding the Feature Fusion model to **3 encoders (adding AST)** improved the greedy score from 0.430 to 0.449. This validates our hypothesis that AST provides complementary global acoustic information. However, combining both ideas, 3 encoders with Learned Resampling, caused performance to drop (0.445). We attribute this to **parameter bloat**: Learned Resampling requires an independent multi-head cross-attention layer for every secondary encoder. Adding AST adds a considerable number of trainable parameters, causing the model to overfit on the small Clotho dataset before it can learn meaningful fusion weights.

Conversely, **Sequence Fusion** proved to be the most resilient strategy. We think this is because it requires the fewest alignment decisions, functioning by simply concatenating projected tokens along the time axis. We attribute its success to three factors: 1) it faithfully preserves all encoder outputs without aggressively squashing or interpolating them; 2) it simplifies the optimization landscape; and 3) it defers the actual cross-modal fusion to the powerful downstream Conformer, essentially stating, “Here are all the tokens; learn the interactions downstream.”

4.3. The Impact of Augmentation and Pre-training

Because our 3-encoder setups were starved for data, we attempted to mitigate overfitting via Stable Audio augmentation. However, greedy scores remained stagnant (0.457 vs. 0.458 for Sequence Fusion). The synthetic domain gap proved too wide to meaningfully improve natural audio translation.

On the other hand, our pre-training configuration using Clotho MixUp and AudioCaps yielded our only other important breakthrough. By pre-training the models, we effectively solved the parameter bloat issue, allowing the heavy 3-encoder fusion layers to properly optimize. This configuration, paired with the highly effective

tive Sequence Fusion, yielded our best overall system, achieving a peak greedy FENSE of 0.486 and a reranked FENSE of **0.531**.

5. CONCLUSION

We presented a configurable multi-encoder architecture for automated audio captioning on Clotho v2. The system combines BEATs with optional ConvNeXt and AST branches, supports feature fusion, sequence fusion, and learned cross-attention resampling, and refines the fused audio representation with a Conformer post-encoder before decoding captions with BART. We also implemented an extended inference pipeline that generates candidate captions with nucleus sampling, re-ranks them using CLAP and decoder likelihood, and consolidates the strongest candidates with an LLM. Our results show that inference-time reranking and LLM-based summarization provide consistent gains over greedy decoding. The best completed configuration reaches a reranked FENSE score of 0.531, compared with its greedy score of 0.486. These improvements suggest that post-processing plays an important role in AAC quality, especially when multiple plausible captions can describe the same audio clip.

This study has several limitations. Not every architectural configuration was fully evaluated with both greedy and reranked inference, and some experiments report only FENSE rather than the full DCASE metric suite. Future work should complete the remaining ablations, evaluate all systems with the full set of captioning metrics, and perform qualitative error analysis to better understand when encoder fusion, augmentation, and LLM summarization help or fail.

6. REFERENCES

- [1] J.-w. Jung, D. Zhang, H. C. H. Yang, S.-L. Wu, D. M. Chan, Z. Kong, D. Ruifan, Z. Yaqian, V. Rafael, and S. Watanabe, "Automatic audio captioning with encoder fusion, multi-layer aggregation, and large language model enriched summarization," DCASE 2024 Challenge, Technical Report, 2024. [Online]. Available: https://dcase.community/documents/challenge2024/technical_reports/DCASE2024_Jung_74_t6.pdf
- [2] DCASE Community, "DCASE 2024 Challenge Task 6: Automated Audio Captioning," <https://dcase.community/challenge2024/task-automated-audio-captioning>, 2024, accessed: 2026-05-10.
- [3] É. Labbé, T. Pellegrini, and J. Pinquier, "CoNeTTE: An efficient audio captioning system leveraging multiple datasets with task embedding," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 3785–3794, 2024.
- [4] S. Chen, Y. Wu, C. Wang, S. Liu, D. Tompkins, Z. Chen, W. Che, X. Yu, and F. Wei, "BEATs: Audio pre-training with acoustic tokenizers," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, ser. Proceedings of Machine Learning Research, vol. 202, 2023.
- [5] T. Pellegrini, I. Khalfaoui-Hassani, É. Labbé, and T. Masquefier, "Adapting a convnext model to audio classification on audioset," 2023.
- [6] Y. Gong, Y.-A. Chung, and J. Glass, "AST: Audio spectrogram transformer," in *Proceedings of Interspeech*, 2021, pp. 571–575.
- [7] K. Drossos, S. Lipping, and T. Virtanen, "Clotho: An audio captioning dataset," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020, pp. 736–740.
- [8] Z. Zhou, Z. Zhang, X. Xu, Z. Xie, M. Wu, and K. Q. Zhu, "Can audio captions be evaluated with image caption metrics?" in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2022, pp. 981–985.
- [9] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.
- [10] W. Chen and X. Xu, "An automated audio captioning system based on large language models," DCASE 2024 Challenge, Technical Report, 2024. [Online]. Available: https://dcase.community/documents/challenge2024/technical_reports/DCASE2024_Chen_2_t6.pdf
- [11] B. Elizalde, S. Deshmukh, M. Al Ismail, and H. Wang, "CLAP: Learning audio concepts from natural language supervision," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- [12] M. Lewis, Y. Liu, N. Goyal, M. Ghazvininejad, A. Mohamed, O. Levy, V. Stoyanov, and L. Zettlemoyer, "BART: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, 2020, pp. 7871–7880.
- [13] A. Gulati, J. Qin, C.-C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu, and R. Pang, "Conformer: Convolution-augmented transformer for speech recognition," in *Proceedings of Interspeech*, 2020, pp. 5036–5040.
- [14] C. D. Kim, B. Kim, H. Lee, and G. Kim, "AudioCaps: Generating captions for audios in the wild," in *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, 2019, pp. 119–132.
- [15] A. Holtzman, J. Buys, L. Du, M. Forbes, and Y. Choi, "The curious case of neural text degeneration," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2020.
- [16] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2019.